

CSCI 135 Assignment – Dictionaries

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October 4th / 6th, 2023

Due Date: Sunday, 10/8/2023 11:59PM

Points: 50

Topic: Dictionaries

The goal of this assignment is to practice using dictionaries in Python. You will use a dictionary to translate text. Continue making your code organized and readable. Pay attention to your header comments and include meaningful comments in-line with your code. Use proper indentation and understandable variable names. Use whitespace to help with readability and logical grouping of sections of code.

Submission:

After following the instructions below, please submit the Python script in Moodle. Upload this as a single file – do not zip it. Be sure the file has a header section including the file name, your name, a description of the program, and other useful information such as example usage.

- ToPirate.py

Grading:

The program/script is worth 50 points. You will be graded according to the following criteria:

	Points
Header Comment	5
Descriptive Comments Throughout Code	5
Program Written Correctly, Following Direction	5
Complete and Correct Dictionary	5
Loops to Collect Input	5
Translates Correctly	5
Stops Correctly	5
Program Runs Correctly, Including Correct Output	15
Total	50

ToPirate.py (aka, Talk Like a Pirate) (aka2, Putting it all together.)

Speaking "Pirate" ... Write a program named ToPirate.py that does the following:

For this assignment, you will write a program that takes interactive input (an English sentence) in the form of a string. The program will then display the Pirate translation of the input English sentence. Use the table below to construct a translation dictionary. You must use a Python dictionary data type (dict) for the translation dictionary.

Topic	English	Pirate Translation
Greetings	hello	avast
	excuse	arrr
People	sir, boy, man	matey
	madam	proud beauty
	officer	foul blaggart
Articles	the	th'
	my	me
	your	yer
	is	be
	are	be
Places	restroom	head
	restaurant	galley
	hotel	fleabag inn

Your program should run and repeatedly ask for English sentences until the user types quit (case insensitive, Quit, QUIT, or similar.) Make sure your check for this also ignores leading and trailing spaces.

Assume the sentences consist of words separated by spaces with no punctuation.

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Following is a test run of the program. User input is shown underlined for clarity. It is not expected that this text be underlined when running your program.

Welcome to the English to Pirate translation service!

When prompted, please enter a sentence in English and it will be translated to Pirate. Beware: the pirate translator's vocabulary is limited, so try not to upset the pirate translator by using fancy words.

Please enter an English sentence:
Excuse me sir where is the restroom

Pirate Translation:
Arrr me matey where be th' head

Please enter an English sentence:
Hello madam would you care to join me for dinner in the restaurant

Pirate Translation:
Avast proud beauty would you care to join me for dinner in th' galley

Please enter an English sentence:
HELLO Madam would YOU care to join me for dinner in The restaurant

Pirate Translation:
Avast proud beauty would you care to join me for dinner in th' galley

Please enter an English sentence:
quit

Thank ye, matey. Ahoy!

At this point, the program terminated because the user entered *quit*.

When an English word is encountered that is not in the the English-to-Pirate dictionary, you should output that word as-is.

Be sure to capitalize the first word!

Pay close attention to other formatting details.

Helpful hints:

How do I get started?

There is a lot to this program. Here are some suggestions:

- Read the problem carefully and write down what steps you need to do.
- Work incrementally.
 - Identify tasks that need to be accomplished to write the program.
 - Identify needed data types and data structures.
 - Identify necessary methods and constructs needed in the code.
 - Try things in the Python interactive console to see that they work.
 - Write code to do pieces of the problem in order and test the pieces as you go.
 - Print out intermediate results and make sure they are correct before moving on.
- Check carefully at the end to ensure that you have covered everything requested.
 - Check again the assignment details.
 - Test thoroughly and carefully.

Upload ToPirate.py to Moodle as your submission for this assignment.

EXTENSIONS

You will not submit the answers to these extensions as part of your assignment submission. You should work through and understand them to learn more and practice. You may be held responsible for the content within, so you really should work through them ...

From the Liang book - NOTE: We will **not** use MyProgrammingLab in the book. You should be able to do exercises from Liang without MyProgrammingLab.

1. *In Section 14.3.1, Liang creates a set using a comprehension. How did he do this? Could he have directly used set comprehension? How could “comprehension” be used to create a tuple containing the same elements?*
2. *In Section 14.3.1, Liang creates a set from a string. The string has 4 characters, but the set has three elements. Why?*
3. *Look at the equality example for sets in Section 14.3.3. Convince yourself why this works the way it does.*
4. *Work through other exercises and examples related to sets in Section 14.3 – 14.5. Do the easy ones first. Try some of the more challenging ones.*
5. *After reading the explanation of what a dictionary is at the beginning of Section 14.6, identify similarities between dictionaries and lists. What are the differences?*
6. *Both dictionaries and sets are represented using curly braces. How does Python determine which is which? How do you create an empty set and an empty dictionary?*
7. *How is looping through a dictionary like looping through a list? How is it different? What is the key distinction?*
8. *Work through other exercises and examples related to sets in Section 14.6 – 14.7. Do the easy ones first. Try some of the more challenging ones.*

From me:

1. The following link shows an online English to Pirate converter that likely inspired the original version of this assignment in prior classes. Notice that it includes multi-word English phrases. How would your approach change to implement this capability in the program you wrote for this assignment?

<http://nifty.stanford.edu/2004/TalkLikeAPirate/pirate2.html>